NAME

frico − Friendly Radiation Integral COde

SYNOPSIS

frico [*OPTION*]...

DESCRIPTION

A 3D Method of Moments code for the simulation of antennas and scattering structures.

This code implements the Method of Moments (MoM), mostly as described in "NASA Contractor Report 189594" titled "MOM3D Method of Moments Code - Theory Manual" by John F. Shaeffer.

The code takes as input a GMSH .geo file and some parameters specifying the sources and the simulation frequency. The output is written to text files (for impedance and radiation patterns) and SILO files for the visualization of fields using VisIt.

OPTIONS

Frico has many command line options and in the following are grouped together in more or less related categories.

Always-necessary options

−g <path>

Path to the input geometry, it must be a GMSH .geo file.

−f <frequency>

Simulation frequency specified in Hz. Exponential notation may be used. Mutually exclusive with -R.

−R <start>:<step>:<end>

Do a sweep from <start> Hz to <end> Hz at steps of <step> Hz. Frequencies may be specified using exponential notation. Mutually exclusive with -f.

Analysis type and related parameters

−s <physgroup>

Perform an antenna analysis by placing a unitary delta-gap source on the specified Physical Curve. Delta-gap voltage is applied to one or more curves that need to be grouped in a physical group of your GMSH model. For example, if a delta-gap source on curves 4,5,6 and 7 is wanted, you need to put in your GMSH model something like

```
Physical Curve("source", 10) = {4,5,6,7};
```

−Z <value>

Set system impedance for SWR calculations (default: 50 Ohm)

−m Monostatic Radar Cross Section computation.

−b <R>:<theta><phi>

Perform a bistatic RCS analysis of the object positioned in the origin of the coordinate axes. The position of the radar is specified by the three arguments **R**, **theta** and **phi**. **R** is the distance from the origin in meters, whereas **theta** and **phi** are the angles in degrees from the pole and around the equator (see COORDINATE SYSTEMS). By setting angles theta = 0 and phi = 0 the radar lies on the Z axis. The radar always transmits in the direction of the origin. The output of the analysis is written in the ‘bistatic_rcs_N.txt‘ files, where N is the sweep point.

BEM/MOM tuning

−A Do not use quadratures, use approximate matrix as described in the MOM3D theory manual.

−k <order>
Integration order to be used when not in approximate mode (-A). Default is 1.

Postprocessing

−P <plane>:<width>:<height>:<meshsize>
−P <nx>:<ny>:<nz>:<width>:<height>:<meshsize>
Add a plane on which E and H fields will be evaluated. The plane can be specified in a 4-parameters form and in a 6-parameters form, and the -P option can be repeated to add multiple planes. The planes are rectangles of a certain **width** and a certain **height** specified in meters. Each plane results in a mesh of triangles whose size is controlled via the parameter **h**; the field will be evaluated in the vertices of that mesh. In the 4-parameters form the parameter **plane** can assume the values xy, yz and xz, yielding an evaluation plane aligned with the specified coordinate axes. The 6-parameter form is not usable yet.

−p <name>:<x>:<y>:<z>
Add a field probe (not ready yet).

Advanced and Debug options

−d Dump matrices and right hand sides of all the linear systems.

−e Force reorientation of basis functions involved in the delta-gap (see CAVEATS).

−x <list of tags>
List of comma-separated tags corresponding to GMSH surfaces you want to exclude from loading. This is useful when in your mesh you have edges with more than two triangles attached, but the option exists only for convenience. The proper way to do this is to delete surfaces directly in the .geo file, for example:

```
SetFactory("OpenCASCADE");
Cylinder(1) = {0, 0, 0, 1, 0, 0, 0.01, 2*Pi};
Cylinder(2) = {-1, 0, 0, 1, 0, 0, 0.01, 2*Pi};
Coherence;
MeshSize{ PointsOf{ Surface{:}; } } = 0.01;
Mesh 2;
Delete { Volume{1,2}; }
Delete { Surface{3}; }
```

−S Expert stuff: In the current implementation the matrix is slightly nonsymmetric (only for non-approximate variant) because of the handling of the self elements. This flag forces symmetrization by doing 0.5*(Z + Zˆ T). Do not use.

Output options

−O <filename>
Enable HDF5 output and write the fields in the file named **filename**.

−o <filename>
Enable Silo output and write the fields in the file named **filename**.

−t Enable textual output.

−n <name>
Set a descriptive name for the simulation. Default is "default".

CAVEATS

Delta-gap sources can occasionally give some problems. What happens is the following:

- **o** The delta-gap is applied on a curve between two entities with different tags

  In this case it is assumed (this should be confirmed with GMSH folks) that all the triangles in the lower-numbered entity have lower tags than the triangles in the higher-numbered entity. In this case there is no problem, because all the RWG basis functions are oriented lower-to-higher and therefore all the edges in the delta-gap will have coherent orientation.

- **o** The delta-gap is applied on curves embedded into a single entity **AND** the curves do not form a closed curve.

  In this case the triangles on both sides of the curves will be part of the same entity, therefore it is not possible to guarantee a coherent orientation of the involved RWG basis functions. In turn, this makes impossible to know the sign of the voltage to apply to the edges of the delta-gap. The code employs a simple algorithm that checks the orientation of adjacent basis functions, and reorients them if it is incoherent. The algorithm works in most cases, but its robustness w.r.t. real-world geometries has not been thoroughly tested.

- **o** Fancier cases

  Not explicitly supported for now, therefore YMMV. Please file a bug describing your situation.

Be aware of this issue if you get nonsensical results when using delta-gaps. This situation can be easily detected by eye by using the "Label" plot in VisIt to inspect the numbering of the elements across the delta-gap.

USAGE EXAMPLES

In the **share/examples** directory provided in the source tree of frico many usage examples are provided.

SEE ALSO

The required theory to understand this code and MoM in general is contained in

- **o** "Field Computations by Moment Methods" by Roger F. Harrington.
- **o** "NASA Contractor Report 189594" titled "MOM3D Method of Moments Code - Theory Manual" by John F. Shaeffer.
- **o** "The Method of Moments in Electromagnetics" by K4LLA Walton C. Gibson.

You should visit the Friuli Venezia Giulia region in Italy and taste "Frico", which is one of the main regional dishes.

BUGS

Report bugs to <matteo.cicuttin@polito.it>

AUTHOR

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